



WOLLO University

Department of software engineering

Exit multiple choice questions for

object oriented programming

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1. Who invented OOP?

- a) Andrea Ferro
- b) Adele Goldberg
- c) Alan Kay
- d) Dennis Ritchie

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Answer: c

Explanation: Alan Kay invented OOP, Andrea Ferro was a part of SmallTalk Development. Dennis invented C++ and Adele Goldberg was in team to develop SmallTalk but Alan actually had got rewarded for OOP.

2. Which is not a feature of OOP in general definitions?

- a) Efficient Code
- b) Code reusability
- c) Modularity
- d) Duplicate/Redundant data

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Answer: d

Explanation: Duplicate/Redundant data is dependent on programmer and hence can't be guaranteed by OOP. Code reusability is done using inheritance. Modularity is supported by using different code files and classes. Codes are more efficient because of features of OOP.

3. Which was the first purely object oriented programming language developed?

- a) Kotlin
- b) SmallTalk
- c) Java
- d) C++

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Answer: b

Explanation: SmallTalk was the first programming language developed which was purely object oriented. It was developed by Alan Kay. OOP concept came into the picture in 1970's.

4. When OOP concept did first came into picture?

- a) 1980's
- b) 1995
- c) 1970's
- d) 1993

[View Answer](#)

Answer: c

Explanation: OOP first came into picture in 1970's by Alan and his team. Later it was used by some programming languages and got implemented successfully, SmallTalk was first language to use pure OOP and followed all rules strictly.

5. Which feature of OOP indicates code reusability?

- a) Abstraction
- b) Polymorphism
- c) Encapsulation
- d) Inheritance

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Answer: d

Explanation: Inheritance indicates the code reusability. Encapsulation and abstraction are meant to hide/group data into one element. Polymorphism is to indicate different tasks performed by a single entity.

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6. Which header file is required in C++ to use OOP?

- a) OOP can be used without using any header file
- b) `stdlib.h`
- c) `iostream.h`
- d) `stdio.h`

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Answer: a

Explanation: We need not include any specific header file to use OOP concept in C++, only specific functions used in code need their respective header files to be included or classes should be defined if needed.

7. Why Java is Partially OOP language?

- a) It allows code to be written outside classes
- b) It supports usual declaration of primitive data types
- c) It does not support pointers
- d) It doesn't support all types of inheritance

[View Answer](#)

Answer: b

Explanation: As Java supports usual declaration of data variables, it is partial implementation of OOP. Because according to rules of OOP, object constructors must be used, even for declaration of variables.

8. Which among the following doesn't come under OOP concept?

- a) Data hiding
- b) Message passing
- c) Platform independent
- d) Data binding

[View Answer](#)

Answer: c

Explanation: Platform independence is not feature of OOP. C++ supports OOP but it's not a platform independent language. Platform independence depends on the programming language.

9. Which is the correct syntax of inheritance?

- a) class base_classname : access derived_classname { /*define class body*/ };
- b) class derived_classname : access base_classname { /*define class body*/ };
- c) class derived_classname : base_classname { /*define class body*/ };
- d) class base_classname : derived_classname { /*define class body*/ };

[View Answer](#)

Answer: b

Explanation: Firstly, keyword class should come, followed by the derived class name. Colon is must followed by access in which base class has to be derived, followed by the base class name. And finally the body of class. Semicolon after the body is also must.

10. Which feature of OOP is indicated by the following code?

```
class student { int marks; };  
class topper:public student { int age; topper(int age){ this.age=age; } };
```

- a) Encapsulation and Inheritance
- b) Inheritance and polymorphism
- c) Polymorphism
- d) Inheritance

[View Answer](#)

Answer: a

Explanation: Encapsulation is indicated by use of classes. Inheritance is shown by inheriting the student class into topper class. Polymorphism is not shown here because we have defined the constructor in the topper class but that doesn't mean that default constructor is overloaded.

11. The feature by which one object can interact with another object is _____

- a) Message reading
- b) Message Passing
- c) Data transfer
- d) Data Binding

[View Answer](#)

Answer: b

Explanation: The interaction between two object is called the message passing feature. Data transfer is not a feature of OOP. Also, message reading is not a feature of OOP.

12. Which among the following, for a pure OOP language, is true?

- a) The language should follow at least 1 feature of OOP
- b) The language must follow only 3 features of OOP
- c) The language must follow all the rules of OOP
- d) The language should follow 3 or more features of OOP

[View Answer](#)

Answer: c

Explanation: The language must follow all the rules of OOP to be called a purely OOP language. Even if a single OOP feature is not followed, then it's known to be a partially OOP language.

13. How many types of access specifiers are provided in OOP (C++)?

- a) 4
- b) 3
- c) 2
- d) 1

[View Answer](#)

Answer: b

Explanation: Only 3 types of access specifiers are available. Namely, private, protected and public. All these three can be used according to the need of security of members.

14. In multilevel inheritance, which is the most significant feature of OOP used?

- a) Code efficiency
- b) Code readability
- c) Flexibility
- d) Code reusability

[View Answer](#)

Answer: d

Explanation: The classes using multilevel inheritance will use the code in all the subsequent subclasses if available. Hence the most significant feature among the options given is code reusability. This feature is generally intended to use the data values and reuse the redundant functions.

15. What is encapsulation in OOP?

- a) It is a way of combining various data members and member functions that operate on those data members into a single unit
- b) It is a way of combining various data members and member functions into a single unit which can operate on any data
- c) It is a way of combining various data members into a single unit
- d) It is a way of combining various member functions into a single unit

[View Answer](#)

Answer: a

Explanation: It is a way of combining both data members and member functions, which operate on those data members, into a single unit. We call it a class in OOP generally. This feature have helped us modify the structures used in C language to be upgraded into class in C++ and other languages.

16. Which of the following is not true about polymorphism?

- a) Helps in redefining the same functionality
- b) Increases overhead of function definition always
- c) It is feature of OOP
- d) Ease in readability of program

[View Answer](#)

Answer: b

Explanation: It never increases function definition overhead, one way or another if you don't use polymorphism, you will use the definition in some other way, so it actually helps to write efficient codes.

17. Which constructor will be called from the object created in the below C++ code?

```
class A
{
    int i;
```

```
A()
{
    i=0; cout<<i;
}
A(int x=0)
{
    i=x; cout<<i;
}
};
A obj1;
```

- a) Parameterized constructor
- b) Default constructor
- c) Run time error
- d) Compile time error

[View Answer](#)

Answer: d

Explanation: When a default constructor is defined and another constructor with 1 default value argument is defined, creating object without parameter will create ambiguity for the compiler. The compiler won't be able to decide which constructor should be called, hence compile time error.

18. What is an abstraction in object-oriented programming?

- a) Hiding the implementation and showing only the features
- b) Hiding the important data
- c) Hiding the implementation
- d) Showing the important data

[View Answer](#)

Answer: a

Explanation: It includes hiding the implementation part and showing only the required data and features to the user. It is done to hide the implementation complexity and details from the user. And to provide a good interface in programming.

19. Which among the following can show polymorphism?

- a) Overloading &&
- b) Overloading <<
- c) Overloading ||
- d) Overloading +=

[View Answer](#)

Answer: b

Explanation: Only insertion operator can be overloaded among all the given options. And the polymorphism can be illustrated here only if any of these is applicable of being overloaded. Overloading is type of polymorphism.

20. In which access should a constructor be defined, so that object of the class can be created in any function?

- a) Any access specifier will work
- b) Private
- c) Public
- d) Protected

[View Answer](#)

Answer: c

Explanation: Constructor function should be available to all the parts of program where the object is to be created. Hence it is advised to define it in public access, so that any other function is able to create objects.

21. Which among the following is correct for the class defined below?

```
class student
{
    int marks;
    public: student(){}
    student(int x)
```

```
{
    marks=x;
}
};
main()
{
    student s1(100);
    student s2();
    student s3=100;
    return 0;
}
```

- a) Program will give compile time error
- b) Object s3, syntax error
- c) Only object s1 and s2 will be created
- d) Program runs and all objects are created

[View Answer](#)

Answer: d

Explanation: It is a special case of constructor with only 1 argument. While calling a constructor with one argument, you are actually implicitly creating a conversion from the argument type to the type of class. Hence you can directly specify the value of that one argument with assignment operator.

22. The copy constructors can be used to _____

- a) Copy an object so that it can be passed to another primitive type variable
- b) Copy an object for type casting
- c) Copy an object so that it can be passed to a function
- d) Copy an object so that it can be passed to a class

[View Answer](#)

Answer: c

Explanation: When an object is passed to a function, actually its copy is made in the function. To copy the values, copy constructor is used. Hence the object being passed and object being used in function are different.

23. Which constructor will be called from the object obj2 in the following C++ program?

```
class A
{
    int i;
    A()
    {
        i=0;
    }
    A(int x)
    {
        i=x+1;
    }
    A(int y, int x)
    {
        i=x+y;
    }
};
A obj1(10);
A obj2(10,20);
A obj3;
```

- a) A(int y, int x)
- b) A(int y; int x)
- c) A(int y)
- d) A(int x)

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Answer: a

Explanation: The two argument constructor will be called as we are passing 2 arguments to the object while creation. The arguments will be passed together and hence compiler resolves that two argument constructor have to be called.

24. Which among the following represents correct constructor?

- a) -classname()
- b) classname()
- c) ()classname
- d) ~classname()

[View Answer](#)

Answer: b

Explanation: The constructors must contain only the class name. The class name is followed by the blank parenthesis or we can have parameters if some values are to be passed.

25. What happens when an object is passed by reference?

- a) Destructor is called at end of function
- b) Destructor is called when called explicitly
- c) Destructor is not called
- d) Destructor is called when function is out of scope

[View Answer](#)

Answer: c

Explanation: The destructor is never called in this situation. The concept is that when an object is passed by reference to the function, the constructor is not called, but only the main object will be used. Hence no destructor will be called at end of function.

26. Which access specifier is usually used for data members of a class?

- a) Protected
- b) Private
- c) Public
- d) Default

[View Answer](#)

Answer: b

Explanation: All the data members should be made private to ensure the highest security of data. In special cases we can use public or protected access, but it is advised to keep the data members private always.

27. How to access data members of a class?

- a) Dot, arrow or direct call
- b) Dot operator
- c) Arrow operator
- d) Dot or arrow as required

[View Answer](#)

Answer: d

Explanation: The data members can never be called directly. Dot operator is used to access the members with help of object of class. Arrow is usually used if pointers are used.

28. Which feature of OOP reduces the use of nested classes?

- a) Inheritance
- b) Binding
- c) Abstraction
- d) Encapsulation

[View Answer](#)

Answer: a

Explanation: Using inheritance we can have the security of the class being inherited. The subclass can access the members of parent class. And have more feature than a nested class being used.

29. Which keyword among the following can be used to declare an array of objects in java?

- a) allocate
- b) arr
- c) new

d) create

[View Answer](#)

Answer: c

Explanation: The keyword new can be used to declare an array of objects in java. The syntax must be specified with an object pointer which is assigned with a memory space containing the required number of object space. Even initialization can be done directly.

30. Which operator can be used to free the memory allocated for an object in C++?

a) Unallocate

b) Free()

c) Collect

d) delete

[View Answer](#)

Answer: d

Explanation: The delete operator in C++ can be used to free the memory and resources held by an object. The function can be called explicitly whenever required. In C++ memory management must be done by the programmer. There is no automatic memory management in C++.

31. Which of the following is not a property of an object?

a) Properties

b) Names

c) Identity

d) Attributes

[View Answer](#)

Answer: b

Explanation: The names are not property of an object. The identity can be in any form like address or name of object but name can't be termed as only identity of an object. The objects contain attributes that define what type of data an object can store.

32. Which type of members can't be accessed in derived classes of a base class?

a) All can be accessed

b) Protected

c) Private

d) Public

[View Answer](#)

Answer: c

Explanation: The private members can be accessed only inside the base class. If the class is derived by other classes. Those members will not be accessible. This concept of OOP is made to make the members more secure.

33. Which among the following best describes the Inheritance?

a) Using the data and functions into derived segment

b) Using already defined functions in a programming language

c) Using the code already written once

d) Copying the code already written

[View Answer](#)

Answer: a

Explanation: It can only be indicated by using the data and functions that we use in derived class, being provided by parent class. Copying code is nowhere similar to this concept, also using the code already written is same as copying. Using already defined functions is not inheritance as we are not adding any of our own features.

34. Single level inheritance supports _____ inheritance.

a) Language independency

b) Multiple inheritance

c) Compile time

d) Runtime

[View Answer](#)

Answer: d

Explanation: The runtime inheritance is done when object of a class is created to call a method. At runtime the

function is searched if it is in class of object. If not, it will search in its parent classes and hierarchy for that method.

35. How to overcome diamond problem?

- a) Using separate derived class
- b) Using virtual keyword with same name function
- c) Can't be done
- d) Using alias name

[View Answer](#)

Answer: b

Explanation: To overcome the ambiguity and conflict we can use keyword virtual. This will help us to differentiate the functions with same name that came to last derived class in diamond problem.

36. Which keyword is used to declare virtual functions?

- a) virt
- b) virtually
- c) virtual
- d) anonymous

[View Answer](#)

Answer: c

Explanation: The virtual keyword is used to declare virtual functions. Anonymous keyword is used with classes and have a different meaning. The virtual functions are used to call the intended function of the derived class.

37. What happens if non static members are used in static member function?

- a) Executes fine
- b) Compile time error
- c) Executes if that member function is not used
- d) Runtime error

[View Answer](#)

Answer: b

Explanation: There must be specific memory space allocated for the data members before the static member functions uses them. But the space is not reserved if object is not declared. Hence only if static members are not used, it leads to compile time error.

38. What is friend member functions in C++?

- a) Non-member functions which have access to all the members (including private) of a class
- b) Member function which doesn't have access to private members
- c) Member function which can modify any data of a class
- d) Member function which can access all the members of a class

[View Answer](#)

Answer: a

Explanation: A non-member function of a class which can access even the private data of a class is a friend function. It is an exception on access to private members outside the class. It is sometimes considered as a member functions since it has all the access that a member function in general have.

39. Where is the memory allocated for the objects?

- a) Cache
- b) ROM
- c) HDD
- d) RAM

[View Answer](#)

Answer: d

Explanation: The memory for the objects or any other data is allocated in RAM initially. This is while we run a program all the memory allocation takes place in some RAM segments. Arrays in heap and local members in stack etc.

40. Which of the following best describes member function overriding?

- a) Member functions having the same name in derived class only
- b) Member functions having the same name and different signature inside main function
- c) Member functions having the same name in base and derived classes

d) Member functions having the same name in base class only

[View Answer](#)

Answer: c

Explanation: The member function which is defined in base class and again in the derived class, is overridden by the definition given in the derived class. This is because the preference is given more to the local members. When derived class object calls that function, definition from the derived class is used.

41. Encapsulation and abstraction differ as _____

- a) Hiding and hiding respectively
- b) Binding and Hiding respectively
- c) Hiding and Binding respectively
- d) Can be used any way

[View Answer](#)

Answer: b

Explanation: Abstraction is hiding the complex code. For example, we directly use cout object in C++ but we don't know how is it actually implemented. Encapsulation is data binding, as in, we try to combine a similar type of data and functions together.

42. Which feature of OOP is exhibited by the function overriding?

- a) Polymorphism
- b) Encapsulation
- c) Abstraction
- d) Inheritance

[View Answer](#)

Answer: a

Explanation: The polymorphism feature is exhibited by function overriding. Polymorphism is the feature which basically defines that same named functions can have more than one functionalities.

43. How to access the private member function of a class?

- a) Using class address
- b) Using object of class
- c) Using object pointer
- d) Using address of member function

[View Answer](#)

Answer: d

Explanation: Even the private member functions can be called outside the class. This is possible if address of the function is known. We can use the address to call the function outside the class.

44. Which keyword should be used to declare static variables?

- a) const
- b) common
- c) static
- d) stat

[View Answer](#)

Answer: c

Explanation: The keyword used to declare static variables is static. This is must be used while declaring the static variables. The compiler can make variables static if and only if they are mentioned with static keyword.

45. Which is correct syntax for declaring pointer to object?

- a) *className objectName;
- b) className* objectName;
- c) className objectName();
- d) className objectName;

[View Answer](#)

Answer: b

Explanation: The syntax must contain * symbol after the className as the type of object. This declares an object pointer. This can store address of any object of the specified class.

46. Which class/set of classes can illustrate polymorphism in the following C++ code?

```
abstract class student
{
    public : int marks;
    calc_grade();
}
class topper:public student
{
    public : calc_grade()
    {
        return 10;
    }
};
class average:public student
{
    public : calc_grade()
    {
        return 20;
    }
};
class failed{ int marks; };
```

- a) Only class student and topper together can show polymorphism
- b) Only class student can show polymorphism
- c) Class failed should also inherit class student for this code to work for polymorphism
- d) All class student, topper and average together can show polymorphism

[View Answer](#)

Answer: d

Explanation: Since Student class is abstract class and class topper and average are inheriting student, class topper and average must define the function named calc_grade(); in abstract class. Since both the definition are different in those classes, calc_grade() will work in different way for same input from different objects. Hence it shows polymorphism.

47. If data members are private, what can we do to access them from the class object?

- a) Private data members can never be accessed from outside the class
- b) Create public member functions to access those data members
- c) Create private member functions to access those data members
- d) Create protected member functions to access those data members

[View Answer](#)

Answer: b

Explanation: We can define public member functions to access those private data members and get their value for use or alteration. They can't be accessed directly but is possible to be access using member functions. This is done to ensure that the private data doesn't get modified accidentally.

48. Which among the following is not a necessary condition for constructors?

- a) Its name must be same as that of class
- b) It must not have any return type
- c) It must contain a definition body
- d) It can contains arguments

[View Answer](#)

Answer: c

Explanation: Constructors are predefined implicitly, even if the programmer doesn't define any of them. Even if the programmer declares a constructor, it's not necessary that it must contain some definition.

49. Object being passed to a copy constructor _____

- a) Must not be mentioned in parameter list
- b) Must be passed with integer type
- c) Must be passed by value
- d) Must be passed by reference

[View Answer](#)

Answer: d

Explanation: This is mandatory to pass the object by reference. Otherwise, the object will try to create another object to copy its values, in turn a constructor will be called, and this will keep on calling itself. This will cause the compiler to give out of memory error.

50. If in multiple inheritance, class C inherits class B, and Class B inherits class A. In which sequence are their destructors called if an object of class C was declared?

- a) ~A() then ~B() then ~C()
- b) ~C() then ~A() then ~B()
- c) ~C() then ~B() then ~A()
- d) ~B() then ~C() then ~A()

[View Answer](#)

Answer: c

Explanation: The destructors are always called in the reverse order of how the constructors were called. Here class A constructor would have been created first if Class C object is declared. Hence class A destructor is called at last.

51. Instance of which type of class can't be created?

- a) Parent class
- b) Abstract class
- c) Anonymous class
- d) Nested class

[View Answer](#)

Answer: b

Explanation: Instance of abstract class can't be created as it will not have any constructor of its own, hence while creating an instance of class, it can't initialize the object members. Actually the class inheriting the abstract class can have its instance because it will have implementation of all members.

52. _____ underlines the feature of Polymorphism in a class.

- a) Virtual Function
- b) Inline function
- c) Enclosing class
- d) Nested class

[View Answer](#)

Answer: a

Explanation: Virtual Functions can be defined in any class using the keyword virtual. All the classes which inherit the class containing the virtual function, define the virtual function as required. Redefining the function on all the derived classes according to class and use represents polymorphism.

53. Which feature in OOP is used to allocate additional functions to a predefined operator in any language?

- a) Function Overloading
- b) Function Overriding
- c) Operator Overloading
- d) Operator Overriding

[View Answer](#)

Answer: c

Explanation: The feature is operator overloading. There is not a feature named operator overriding specifically. Function overloading and overriding doesn't give addition function to any operator.

54. Which feature can be implemented using encapsulation?

- a) Polymorphism
- b) Overloading
- c) Inheritance
- d) Abstraction

[View Answer](#)

Answer: d

Explanation: Data abstraction can be achieved by using encapsulation. We can hide the operation and structure of actual program from the user and can show only required information by the user.

What is the implicit return type of constructor?

No return type

A class object in which it is defined.

void

None

Hide

Wrong Answer

Explanation - B) Implicit return type of constructor is the class object in which it is defined.

2.

When is the object created with a new keyword?

At run time

At compile time

Depends on the code

None

Hide

Wrong Answer

Explanation - A) The object created with a new keyword during run-time.

3.

Identify the incorrect constructor type.

Friend constructor

Default constructor

Parameterized constructor

Copy constructor

Hide

Wrong Answer

Explanation - A) Friend constructor is not a type of constructor.

4.

Identify the scope resolution operator.

:

::

?:

None

Hide

Wrong Answer

Answer - B) :: is the scope resolution operator.

5.

Choose the option below which is not a member of the class.

Friend function

Static function

Virtual function

Const function

Hide

Wrong Answer

Answer - A) Friend function is not a member of the class.

6.

Total types of constructors in C++ are?

1

2

3

4

Hide

Wrong Answer

Answer - C) There are three types of constructors in C++ - .default constructor, parameterized constructor and copy constructor.

7.

What is the number of parameters that a default constructor requires?

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0

1

2

3

Hide

Wrong Answer

Answer - A) Default constructor does not require any parameter.

8.

Data members and member functions of a class are private. default. True or False?

True

False

Depends on code

None

Hide

Wrong Answer

Answer - A) True. Data members and member functions of a class are always private by default.

9.

Under which pillar of OOPS do base class and derived class relationships come?

Polymorphism

Inheritance

Encapsulation

Abstraction

Hide

Wrong Answer

Answer - B) Base class and derived class relationship come under inheritance.

10.

Which of the following functions can be inherited from the base class?

Constructor

Destructor

Static

None

Hide

Wrong Answer

Answer - D) None of the functions can be inherited from the base class.

11.

Which of the following is not a type of inheritance?

Multiple

Multilevel

Distributed

Hierarchical

Hide

Wrong Answer

Answer - C) Distributed is not a type of inheritance.

12.

What is an object in c++?

It is a function of class

It is an instance of the class

It is the data type of class

It is part of the syntax of class.

Hide

Wrong Answer

Answer - B) An object is an instance of the class.

13.

Why is reusability a desirable feature?

Reduces compilation time

Decreases testing time

Lowers maintenance cost

None

Hide

Wrong Answer

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Answer - B) Reusability is a desirable feature as it decreases the testing time.

14.

Identify the operators which cannot be overloaded.

?:

.(dot operator)

>>

Both A and B

Hide

Wrong Answer

Answer - C) >> cannot be overloaded.

15.

Another name of overloading is?

Pseudo polymorphism

Transient polymorphism

Virtual polymorphism

Ad-hoc polymorphism

Hide

Wrong Answer

Answer - D) Overloading is also known as Ad-hoc polymorphism.

16.

By default, fields in a structure of a C program is?

public

Private

protected

None

Hide

Wrong Answer

Answer - A) Fields in the structure are public by default

17.

On what basis is it determined, when a variable comes into existence in memory?

Data type

Storage class

Scope

All of the above

Hide

Wrong Answer

Answer - B) A variable that comes into existence in memory is determined by storage class.

18.

Identify the feature using which, one object can interact with another object.

Message passing

Message reading

Data binding

Data transfer

Hide

Wrong Answer

Answer - A) Message passing is the feature using which, one object can interact with another object

19.

Total access specifiers in OOPS for C++ are?

1

2

3

4

Hide

Wrong Answer

Answer - C) There is a total of 3 access specifiers - private, public, protected.

20.

Choose the option below which can show polymorphism.

Overloading ||

Overloading <<

Overloading &&

Overloading +=

Hide

Wrong Answer

Answer - B) Overloading << can show polymorphism.

21.

Select the following which shows the correct constructor.

()class_name

-class_name

class_name()

~class_name()

Hide

Wrong Answer

Answer - C) class_name() is the correct constructor.

22.

To access data members of a class, which of the following is used?

Dot operator

Arrow operator

Dot or arrow operator, as required

Dot, arrow, or direct call

Hide

Wrong Answer

Answer - C) To access data members of a class, Dot or arrow operator, as required.

23.

Identify the feature, which is used to reduce the use of nested classes.

Binding

Inheritance

Encapsulation

Abstraction

Hide

Wrong Answer

Answer - B) Inheritance is the feature. which is used to reduce the use of nested classes.

24.

Choose the option below which is used to free the memory allocated for an object in C++.

Free()

unallocate()

delete

collect

Hide

Wrong Answer

Answer - C) delete is used to free the memory allocated for an object in C++.

25.

Identify the option below which is not a property of the object.

Names

Attributes

Identity

Properties

Hide

Wrong Answer

Answer - A) Names are not a property of the object.

26.

What type of inheritance does single-level inheritance support?

Compile-time

Runtime inheritance

Multiple inheritances

Language independent

Hide

Wrong Answer

Answer - B) Single-level inheritance supports Runtime inheritance.

27.

In which of the following is memory allocated for the objects?

RAM

ROM

Cache

HDD

Hide

Wrong Answer

Answer - A) Memory is allocated for the objects in RAM

28.

How do encapsulation and abstraction differ?

Hiding and binding

Binding and hiding

Hiding and hiding

None

Hide

Wrong Answer

Answer - B) Encapsulation and abstraction differ on the basis of binding and hiding.

29.

Choose the option below which is shown by function overriding

Abstraction

Encapsulation

Polymorphism

Inheritance

Hide

Wrong Answer

Answer - C). Function overriding shows polymorphism.

30.

Choose the option below for which instance of the class cannot be created.

Anonymous class

Parent class

Nested class

Abstract class

Hide

Wrong Answer

Answer - D) For the Abstract class, an instance of a class cannot be created.

31.

Identify the feature which can be implemented using encapsulation.

Abstraction

Encapsulation

Polymorphism

Inheritance

Hide

Wrong Answer

Answer - A) Abstraction can be implemented using encapsulation.

32.

Why is a virtual function mainly used to achieve?

Function code polymorphism

Interpreter polymorphism

Compile-time polymorphism

Runtime polymorphism

Hide

Wrong Answer

Answer - D) Virtual function mainly used to achieve Runtime polymorphism.

33.

Virtual functions should be defined in?

Derived class

Base class

Both base and derived class

None

Hide

Wrong Answer

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Answer - B) Virtual functions should be defined in the base class.

34.

Virtual function should be _____.

public

Private

protected

default

Hide

Wrong Answer

Answer - A) Virtual functions should be public.

35.

Choose whether True or False: The constructor function can be constant.

True

False

Depends on implementation

None

Hide

Wrong Answer

Answer - B) False. Constructor function cannot be constant.

36.

Dynamic memory allocation can be done using?

calloc()

malloc()

Both (a) and (b)

create()

Hide

Wrong Answer

Answer - C) Dynamic memory allocation can be done using both calloc() and malloc().

37.

Total types of polymorphism in C++ are?

1

2

3

4

Hide

Wrong Answer

Answer - B) There is a total of 2 types of polymorphism in C++ - runtime polymorphism and compile-time polymorphism.

38.

Another name of the generic class is?

Template class

Final class

Abstract class

Efficient code

Hide

Wrong Answer

Answer - A) Generic class is also known as a template class.

39.

Choose the option below which overloads using the friend function.

()

->

*

=

Hide

Wrong Answer

Answer - C) * can be overloaded using the friend function.

40.

Choose whether True or False: Object cannot be passed as a function.

True

False

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Depends on implementation

None

Hide

Wrong Answer

Answer - A) True. The object cannot be passed as a function.

41.

Identify the pure object-oriented programming language among the following.

C++

SmallTalk

Java

Python

Hide

Wrong Answer

Answer - B) SmallTalk is a pure object-oriented programming language.

42.

Identify the abstract data type among the following.

double

int

class

string

Hide

Wrong Answer

Answer - C) Class is an abstract data type.

43.

What are the total catch blocks that can be used using a single Try block?

1

2

According to needs

Maximum 256

Hide

Wrong Answer

Answer - C) We can use as many catch blocks as required using a single Try block.

44.

Choose the incorrect option below which is not a type of constructor.

Copy constructor

Friend constructor

Parameterized constructor

Default constructor

Hide

Wrong Answer

Answer - B). Friend constructor is not a type of constructor.

45.

Total instances of an abstract class that can be created?

0

1

4

8

Hide

Wrong Answer

Answer - A) Instances of abstract class cannot be made.

46.

Identify the option below which cannot be used with the term “virtual”.

constructor

destructor

class

Member function

Hide

Wrong Answer

Answer - A) Constructor cannot be used with the term “virtual”.

47.

Which type of inheritance is implemented in the following code below?

- Single inheritance
- Multilevel inheritance
- Hybrid inheritance
- Multiple inheritances

Hide

Wrong Answer

Answer - D) Multiple inheritances are implemented in the above code.

48.

Choose the option below which is used to implement late binding.

- Friend function
- Static function
- Virtual function
- Const function

Hide

Wrong Answer

Answer - C) Virtual function is used to implement late binding.

49.

What is cout?

- Operator
- Object
- Function
- Macro

Hide

Wrong Answer

Answer - B) cout is an object.

50.

Choose the option below which describes the overloading of functions.

- Ad-hoc polymorphism
- Virtual polymorphism
- Pseudo polymorphism
- Transient polymorphism

Hide

Wrong Answer

Answer - A) Ad-hoc polymorphism correctly describes the overloading of functions

51.

Who developed object-oriented programming?

- Dennis Ritchie
- Charles Babbage
- Alan Kay
- Andrea Ferro

Hide

Wrong Answer

Answer - C) Object-oriented programming was developed by Alan Kay.

52.

Which of the following is not an oops concept?

- Inheritance
- Compilation
- Polymorphism
- Encapsulation

Hide

Wrong Answer

Answer - B) Compilation is not a concept of OOPS

1) Which of the following language was developed as the first purely object programming language?

- a. SmallTalk
- b. C++
- c. Kotlin
- d. Java

Hide Answer Workspace

Answer: a. SmallTalk

Explanation: This programming language was invented as the first pure OOPS (object-oriented) language. This language was designed by Alan Kay in the early 1970s.

2) Who developed object-oriented programming?

- a. Adele Goldberg
- b. Dennis Ritchie
- c. Alan Kay
- d. Andrea Ferro

Hide Answer Workspace

Answer: c. Alan Kay

Explanation: In the year 1970, Alan Kay gave Object-Oriented programming. He coined the concept of OOPS at a grad school in the year 1966 or 1967. Alan kay, Adele Goldberg, Dan Ingalls and others developed the first Smalltalk programming language, which follows the OOPS concept.

3) Which of the following is not an OOPS concept?

- a. Encapsulation
- b. Polymorphism
- c. Exception
- d. Abstraction

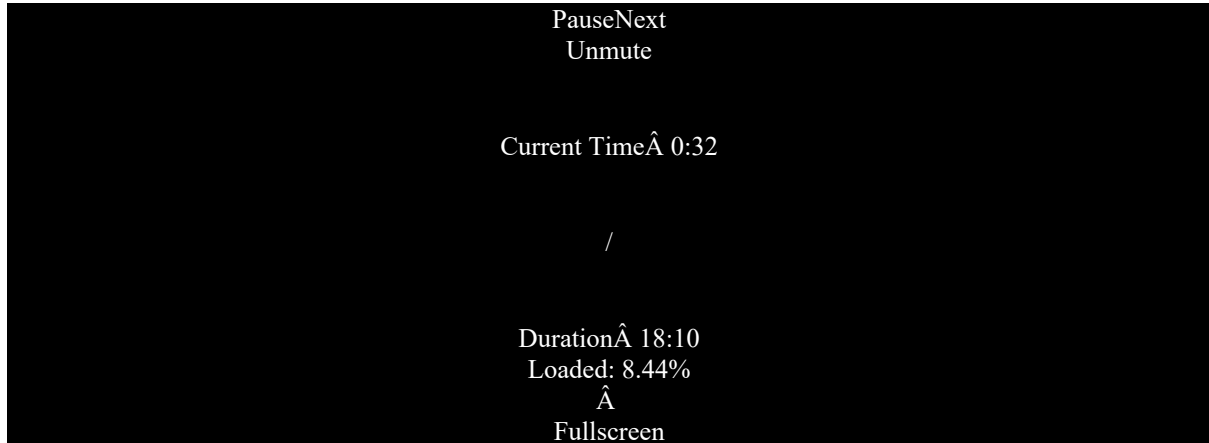
Hide Answer Workspace

Answer: c. Exception

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Explanation: None.

4) Which feature of OOPS described the reusability of code?



- a. Abstraction
- b. Encapsulation
- c. Polymorphism
- d. Inheritance

Hide Answer Workspace

Answer: d. Inheritance

Explanation: Inheritance is the feature of OOPS, which allows the users of OOPS to reuse the code which is already written. This OOPS feature inherits the features of another class in the programs. This mechanism actually inherits the fields and methods of the superclass.

5) Which of the following language supports polymorphism but not the classes?

- a. C++ programming language
- b. Java programming language
- c. Ada programming language
- d. C# programming language

Hide Answer Workspace

Answer: c. Ada programming language

Explanation: It is a programming language that disapproves of the concept of polymorphism but supports the concept of classes. It is an object-based language. So, it does not follow the Object-oriented programming concepts.

6) Which among the following feature is not in the general definition of OOPS?

- a. Modularity
- b. Efficient Code
- c. Code reusability
- d. Duplicate or Redundant Data

Hide Answer Workspace

Answer: d. Duplicate or Redundant Data

Explanation: Duplicacy or Redundancy of data is a feature which totally dependent on the programmers. So, it cannot be created by the OOPS.

7) Which feature of OOPS derives the class from another class?

- a. Inheritance
- b. Data hiding
- c. Encapsulation
- d. Polymorphism

Hide Answer Workspace

Answer: a. Inheritance

Explanation: Inheritance is an important OOPS feature which derives the class from the base class or superclass. This OOPS feature inherits the features of another class in the programs. This mechanism actually inherits the fields and methods of the superclass.

8) Define the programming language, which does not support all four types of inheritance?

- a. Smalltalk
- b. Kotlin
- c. Java
- d. C++

Hide Answer Workspace

Answer: c. Java

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Explanation: Java is a programming language that disapproves of the concept of 'multiple inheritance'. So, it does not agree with all types of inheritance. But, we can implement 'multiple inheritance' in Java language using the interface concept.

9) A single program of OOPS contains _____ classes?

- a. Only 1
- b. Only 999
- c. Only 100
- d. Any number

Hide Answer Workspace

Answer: d. Any number

Explanation: We can define any number of classes with different names in a single program of OOPS.

10) Which operator from the following can be used to illustrate the feature of polymorphism?

- a. Overloading <<
- b. Overloading &&
- c. Overloading ||
- d. Overloading +=

Hide Answer Workspace

Answer: a. Overloading <<

Explanation: << is an insertion operator which is used for overloading (polymorphism).

11) Which two features of object-oriented programming are the same?

- a. Abstraction and Polymorphism features are the same
- b. Inheritance and Encapsulation features are the same
- c. Encapsulation and Polymorphism features are the same
- d. Encapsulation and Abstraction

Hide Answer Workspace

Answer: d. Encapsulation and Abstraction

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Explanation: Encapsulation and Abstraction are the same OOPS concepts. Encapsulation hides the features of the object and binds all the properties inside a single class. And abstraction is a feature that shows the required data to the user.

12) Which header file is required by the C++ programming language to use the OOPS concept?

- a. `stdio.h`
- b. `iostream.h`
- c. `stdlib.h`
- d. We can easily use the OOPS concepts in c++ programs without using any header file.

Hide Answer Workspace

Answer: d. We can easily use the OOPS concepts in c++ programs without using any header file.

Explanation: There is no need to use any particular header file for using the OOPS concept in the C++ programs. The C++ functions and variables have their respective header files, which should be defined in the program.

13) Which of the following definition is incorrect for polymorphism?

- a. Polymorphism helps in redefining the same functionality
- b. Polymorphism concept is the feature of object-oriented programming (OOP)
- c. It always increases the overhead of function definition
- d. Ease in the readability of the program

Hide Answer Workspace

Answer: c. It always increases the overhead of function definition

Explanation: This concept of OOPS never increases the overhead of function definition.

14) Which among the following cannot be used for the concept of polymorphism?

- a. Static member function
- b. Constructor Overloading
- c. Member function overloading
- d. Global member function

Hide Answer Workspace

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Answer: a. Static member function

Explanation: These functions are not an object property. That's why they cannot be acceptable for overriding or overloading.

15) Which function best describe the concept of polymorphism in programming languages?

- a. Class member function
- b. Virtual function
- c. Inline function
- d. Undefined function

Hide Answer Workspace

Answer: b. Virtual function

Explanation: Only those functions are used to achieve the polymorphism, which are declared as 'virtual'. These functions let the OOPS programs decide at runtime which function is to be called by the pointer.

16) Which member function is assumed to call first when there is a case of using function overloading or abstract class?

- a. Global function
- b. Local function
- c. Function with lowest priority
- d. Function with the highest priority

Hide Answer Workspace

Answer: d. Function with the highest priority

Explanation: The member function with the highest priority is called first when there is function overloading, or abstract class is used.

17) Is it true to use polymorphism in the C programming language?

- a. True
- b. False

Hide Answer Workspace

Answer: a. True

Explanation: Yes, we can use the concept of polymorphism in the C programming language. Users can use structures and then declare pointers in C programming language, which in turn points to some function.

Here, he/she can easily simulate the functions which are not exactly member function. Hence, users can manipulate the concept of polymorphism in the C language.

18) Which of the following language uses the classes but not the polymorphism concept?

- a. Procedure Oriented language
- b. Object-based language
- c. Class-based language
- d. If classes are used, then the polymorphism concept will always be used in the programming languages.

Hide Answer Workspace

Answer: b. Object-based language

Explanation: Object-based languages are those languages which support the definition of classes but not the polymorphism. Inheritance is also another feature which is not supported by these languages.

19) Which of the following OOP concept is not true for the C++ programming language?

- a. A class must have member functions
- b. C++ Program can be easily written without the use of classes
- c. At least one instance should be declared within the C++ program
- d. C++ Program must contain at least one class

Hide Answer Workspace

Answer: d. C++ Program must contain at least one class

Explanation: Use of class in the C++ language is not mandatory. Programs can be easily written without the use of classes. C++ programs classes may or may not contain the member function. That's why it is not a necessary situation in the C++ language.

20) What is the extra feature in classes which was not in the structures?

- a. Member functions
- b. Data members

- c. Public access specifier
- d. Static Data allowed

Hide Answer Workspace

Answer: a. Member functions

Explanation: Member function is an extra feature which is allowed in class but not in the concept of structures.

21) How many types of polymorphism in the C++ programming language?

- a. Three types of polymorphism
- b. Two types of polymorphism
- c. Five types of polymorphism
- d. Four types of polymorphism

Hide Answer Workspace

Answer: b. Two types of polymorphism

Explanation: C++ programming language has two types of polymorphism: 1. Runtime Polymorphism 2. Compile-time Polymorphism

Runtime Polymorphism: It is met by the function overriding. This polymorphism is also known as late or dynamic binding.

Compile-time Polymorphism: It is met by the operator and function overloading. This polymorphism is also known as early or static binding.

22) Which of the following feature is also known as run-time binding or late binding?

- a. Dynamic typing
- b. Dynamic loading
- c. Dynamic binding
- d. Data hiding

Hide Answer Workspace

Answer: c. Dynamic binding

Explanation: Dynamic binding or runtime binding or late binding is that type of binding which happens at the execution time of the program or code. Function or method overriding is the perfect example of this type of binding. Virtual functions are used to achieve the concept of function overriding.

23) Which among the following is not a member of the class?

- a. Virtual function
- b. const function
- c. Static function
- d. Friend function

Hide Answer Workspace

Answer: d. Friend function

Explanation: The public member functions of a class can easily access the private data members of the same class.

This is achieved by the "friend", which is a non-member function to the class. Its private data can be accessed. That's why the friend function is not a member of the class.

24) Which of the following class is known as the generic class?

- a. Final class
- b. Template class
- c. Abstract class
- d. Efficient code

Hide Answer Workspace

Answer: b. Template class

Explanation: Template classes are those classes which can be used for any value of data type. So, these are known as a generic class.

Template classes help in making the genetic classes and generate the objects of classes based on the parameters. This type of class also saves system memory.

25) Which operator overloads using the friend function?

- a. *
- b. ()
- c. ->
- d. =

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Hide Answer Workspace

Answer: a. *

Explanation: The operators (->, (), =) cannot be overloaded using the friend function because if they are overloaded, then the code will show the compilation error. That's why * (asterisk) is a symbol that can be overloaded using the friend function.

26) Which of the following OOP concept binds the code and data together and keeps them secure from the outside world?

- a. Polymorphism
- b. Inheritance
- c. Abstraction
- d. Encapsulation

Hide Answer Workspace

Answer: d. Encapsulation

Explanation: Encapsulation is an important concept of Object-oriented programming. This concept binds the data and methods in a single unit. It binds the methods which manipulate the data.

27) Which member of the superclass is never accessible to the subclass?

- a. Public member
- b. Protected member
- c. Private member
- d. All of the mentioned

Hide Answer Workspace

Answer: c. Private member

Explanation: Private is a member of a class which never be accessible to the derived class.

28) What is the size of a class?

- a. Sum of the size of all inherited variables along with the variables of the same class
- b. The size of the class is the largest size of the variable of the same class
- c. Classes in the programming languages do not have any size

- d. Sum of the size of all the variables within a class.

Hide Answer Workspace

Answer: c. Classes in the programming languages do not have any size

Explanation: The object size defines the size of the class. That's why classes do not have any size.

29) Which class cannot create its instance?

- a. Parent class
- b. Nested class
- c. Anonymous class
- d. Abstract class

Hide Answer Workspace

Answer: d. Abstract class

Explanation: The instance of the abstract class can't be created because it will not have any constructor of its own. Hence while creating an instance of a class, it can't initialize the object members.

30) Encapsulation adds the function in a user-defined structure.

- a. True
- b. False

Hide Answer Workspace

Answer: b. False

Explanation: The encapsulation concept does not add the function in a user-defined structure because if the member functions are involved, then the user cannot call the structures defined by it.

31) Which of the following variable violates the definition of encapsulation?

- a. Array variables
- b. Local variables
- c. Global variables
- d. Public variables

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Hide Answer Workspace

Answer: c. Global variables

Explanation: Global variable is a variable which violates the definition of encapsulation. The definition of encapsulation defines that the data or variable should be accessed by the specified set of elements. But, the global variable is accessible everywhere in the code. This type of variable does not hide the internal performance of the code. That's global variable violates the encapsulation concept.

32) How can the concept of encapsulation be achieved in the program?

- a. By using the Access specifiers
- b. By using the concept of Abstraction
- c. By using only private members
- d. By using the concept of Inheritance

Hide Answer Workspace

Answer: a. By using the Access specifiers

Explanation: Users can achieve the concept of encapsulation by implementing the access specifiers in the code. It is not compulsory that the user use only private members.

33) The concept of encapsulation helps in writing which type of classes in the Java programming language?

- a. Abstract classes
- b. Wrapper classes
- c. Mutable classes
- d. Immutable classes

Hide Answer Workspace

Answer: d. Immutable classes

Explanation: The concept of encapsulation helps in writing immutable classes in the Java programming language because these classes are used for caching purposes.

34) Encapsulation is _____?

- a. technique of combining more than one member functions into a single unit.
- b. mechanism of combining more than one data member into a single unit.

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- c. mechanism of combining more than one data members and member functions that implement on those data members into a single unit
- d. technique of combining more than one data members and member functions into a single unit, which can manipulate any data.

Hide Answer Workspace

Answer: c. mechanism of combining more than one data members and member functions that implement on those data members into a single unit

Explanation: Encapsulation is an OOPS mechanism that combines more than one data member and member function and implements the member functions on those data members into a single unit.

35) Which of the following statement of a program is not right?

- a. `class teacher{ }; teacher s[5];`
- b. `class teacher{ }s;`
- c. `class teacher{ }; teacher s;`
- d. `class teacher{ }s[];`

Hide Answer Workspace

Answer: d. `class teacher{ }s[];`

Explanation: This statement is false because it must be compulsory to specify the array with its size.

36) Which of the following syntax is incorrect for the class definition?

- a. `student class{ };`
- b. `class student{ student(int a){} };`
- c. `class teacher{ public: teacher(int a){ } };`
- d. None of the mentioned

Hide Answer Workspace

Answer: a. `student class{ };`

Explanation: In this syntax, the keyword 'class' is present after the student. That's why this syntax is incorrect.

37) The object cannot be _____?

- a. passed by copy
- b. passed as function
- c. passed by value
- d. passed by reference

Hide Answer Workspace

Answer: b. passed as function

Explanation: The object cannot be passed as the function because it is an instance of a class. Only the objects passed by the value, reference or copy.

38) Which among the following feature does not come under the concept of OOPS?

- a. Data binding
- b. Data hiding
- c. Platform independent
- d. Message passing

Hide Answer Workspace

Answer: c. Platform independent

Explanation: Platform independence is a feature which does not come under the OOPS concepts. This feature depends on the programming language. C++ is an object-oriented programming language which is not a platform-independent language.

39) Which of the following feature may be broken if the user does not use the classes in the code?

- a. Object must be used violated
- b. Only the encapsulation concept is violated
- c. Inheritance cannot be implemented
- d. Basically, all the features of OOPS get violated

Hide Answer Workspace

Answer: d. Basically, all the features of OOPS get violated

Explanation: If the user does not use the classes in the code, then all the OOP features get violated. And, the encapsulation and inheritance cannot be implemented without the use of classes.

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40) Which of the following feature interacts one object with another object?

- a. Message reading
- b. Message passing
- c. Data transfer
- d. Data binding

Hide Answer Workspace

Answer: b. Message passing

Explanation: Message passing is a feature of OOP that connects one object with another object.

It is a concept of communication for passing the messages between the two objects. It allows objects to send and receive information among them.

41) Which definition best defines the concept of abstraction?

- a. Hides the important data
- b. Hides the implementation and showing only the features
- c. Hiding the implementation
- d. Showing the important data

Hide Answer Workspace

Answer: b. Hides the implementation and showing only the features

Explanation: It hides the implementation part and showing only the data and features which are required to the end-user.

This technique is used to hide the complexity of code and its details from the user. It is also used to define a good interface in coding.

42) The combination of abstraction of the data and code is viewed in_____.

- a. Inheritance
- b. Object
- c. Class
- d. Interfaces

Hide Answer Workspace

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Answer: b. Object

Explanation: Object is a technique which can be viewed as an abstraction of the combination of data and code. This OOPS concept uses the class data members and their functioning as the data abstraction. Code abstraction is used as the object of an inbuilt class.

43) The principle of abstraction _____

- a. is used to achieve OOPS.
- b. is used to avoid duplication
- c. Use abstraction at its minimum
- d. is used to remove longer codes

Hide Answer Workspace

Answer: b. is used to avoid duplication

Explanation: The principle of abstraction is used to avoid the code duplicacy. In this technique, the program does not contain any redundant functions. And it also makes the program effective.

44) Which among the following concept is correct if a user using the concept of encapsulation in a code?

- a. The modification of the code can be additional overhead
- b. Member functions can be used for modifying the data type of data members
- c. The data type of data member cannot be modified.
- d. The data type of the data member can be easily modified without modifying any other code

Hide Answer Workspace

Answer: d. The data type of the data member can be easily modified without modifying any other code

Explanation: Data type of a data member can be easily modified without modifying any code. Member functions can never modify the data type of the data members in the same class.

45) Using the concept of encapsulation security of the data is _____

- a. Ensured to some extent
- b. Purely ensured
- c. Not ensured
- d. Very low

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Hide Answer Workspace

Answer: a. Ensured to some extent

Explanation: The concept of encapsulation can only ensure data security to some extent.

46) Consider the following Java program and select the right option from the given options.

- ```
1. class marksofstudent
2. {
3. int subjectmarks;
4. public : int* fun()
5. {
6. return &subjectmarks;
7. }
8. };
9. main()
10. {
11. marksofstudent s;
12. int *ptr = c.fun() ;
13. return 0;
14. }
```
- a. The above program violates the feature of encapsulation
  - b. The above program may result in undesirable conditions
  - c. The above program will generate an error
  - d. The above program is good to go

Hide Answer Workspace

**Answer:** a. The above program violates the feature of encapsulation

**Explanation:** The code in this question violates the definition of encapsulation.

---

47) The name of the default access specifier for the member functions or data members in the C++ programming language is\_\_\_\_\_.

- a. Private access specifier
- b. Public access specifier

- c. Protected access specifier
- d. Depends on compiler

Hide Answer Workspace

**Answer:** a. Private access specifier

**Explanation:** If none of the access specifiers is used, then the data members and member functions are Private by default in C++ classes. This specifier is actually used to increase the privacy of information.

---

48) Which of the following option best illustrates a friend class?

- a. This class can access and manipulate all the private members of that class which connects to a friend.
- b. Friend class can only access and manipulate the protected data members of that class that connects to a friend.
- c. Friend class can't access any data member of another class but can use its methods
- d. Friend class don't have any implementation

Hide Answer Workspace

**Answer:** a. Friend class can access and manipulate all the private members of the class, of which it is a friend

**Explanation:** A friend class can easily access all the private data members of another class.

---

49) Which of the following definition best describes the concept of polymorphism?

- a. It is the ability to process the many messages and data in one way
- b. It is the ability to process the undefined messages or data in at least one way
- c. It is the ability to process the message or data in more than one form
- d. It is the ability to process the message or data in only one form

Hide Answer Workspace

**Answer:** c. It is the ability for a message or data to be processed in more than one form.

**Explanation:** None.

---

50) Which class/ or set of classes can describe the concept of polymorphism in the following code?

1. **abstract class** student\_details

```
2. {
3. public : int marksofstudent;
4. calculate_grade();
5. }
6. class topper:public student_details
7. {
8. public : calculate_grade()
9. {
10. return 15;
11. }
12. };
13. class average:public student_details
14. {
15. public : calculate_grade()
16. {
17. return 30;
18. }
19. };
20. class failed{ int marksofstudent; };
```

- a. Only the student\_details class can show the concept of polymorphism
- b. The class which is 'failed' should also inherit class student for this code to work for polymorphism
- c. The student\_details, topper and average classes together can show the concept of polymorphism
- d. Only the student\_details and topper class together can show the concept of polymorphism

Hide Answer Workspace

**Answer:** c. The student\_details, topper and average classes together can show the concept of polymorphism

**Explanation:** None.